

Customer Shopping Behavior Analysis





Description of the project

An end-to-end data project examining 3,900 transactions across product categories to uncover insights on spending, customer segments, product preferences, and subscription behavior — driving strategic business decisions.

Dataset Summary

3900

Transactions

Transactions Total purchase
records analyzed

18

Columns

Features per transaction

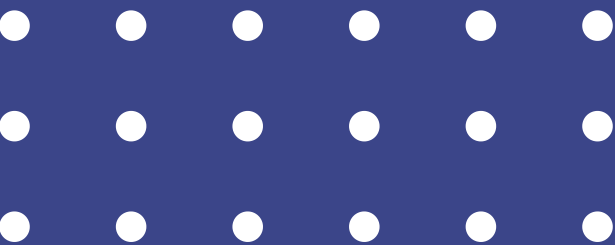
37

Missing Values

In review rating column only

Key Features

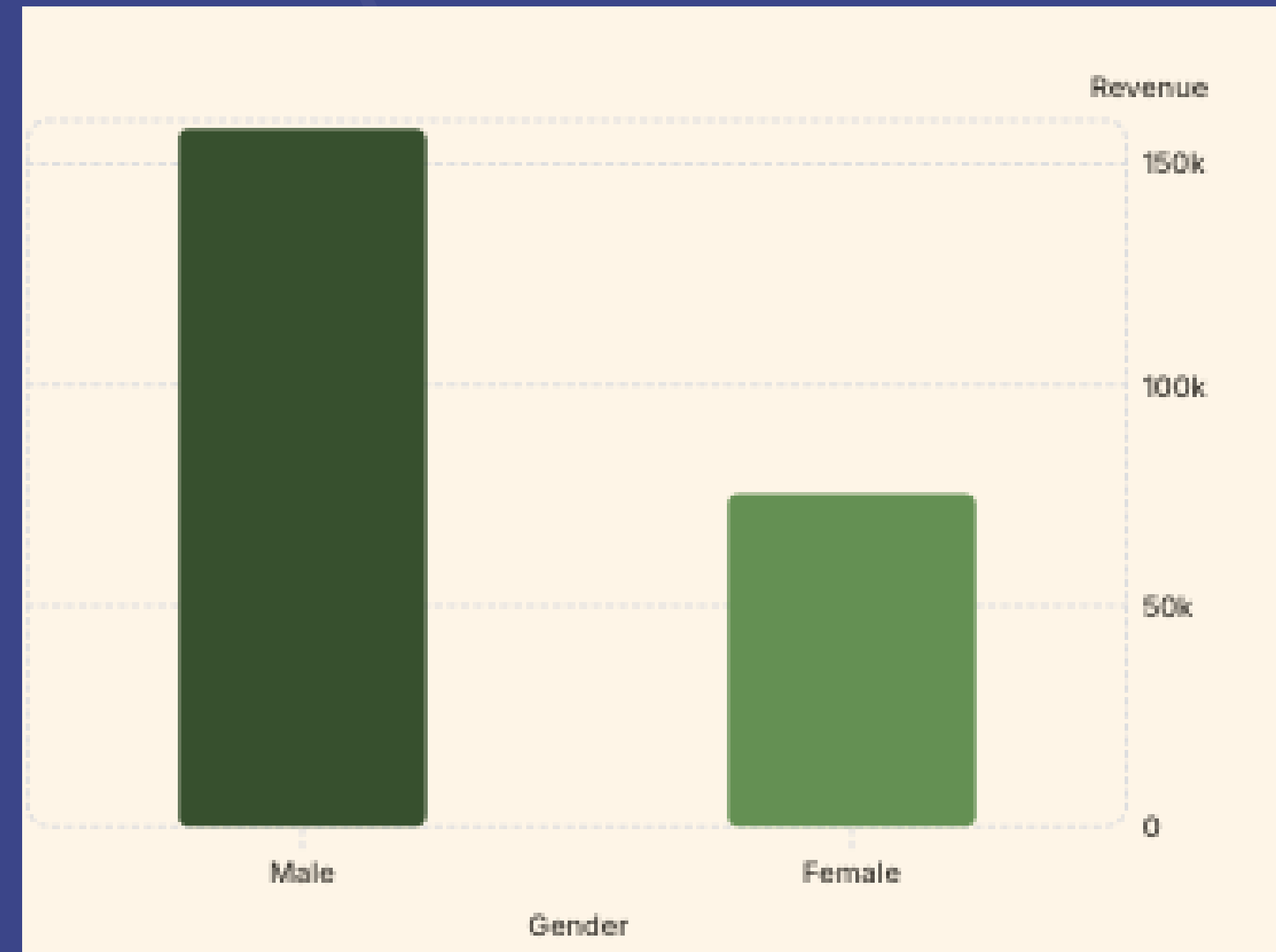
- Demographics: Age, Gender, Location, Subscription Status
- Purchase Details: Item, Category, Amount, Season, Size, Color
- Behavior: Discount, Promo Code, Frequency, Review Rating, Shipping Type



Revenue: Male vs. Female Customers

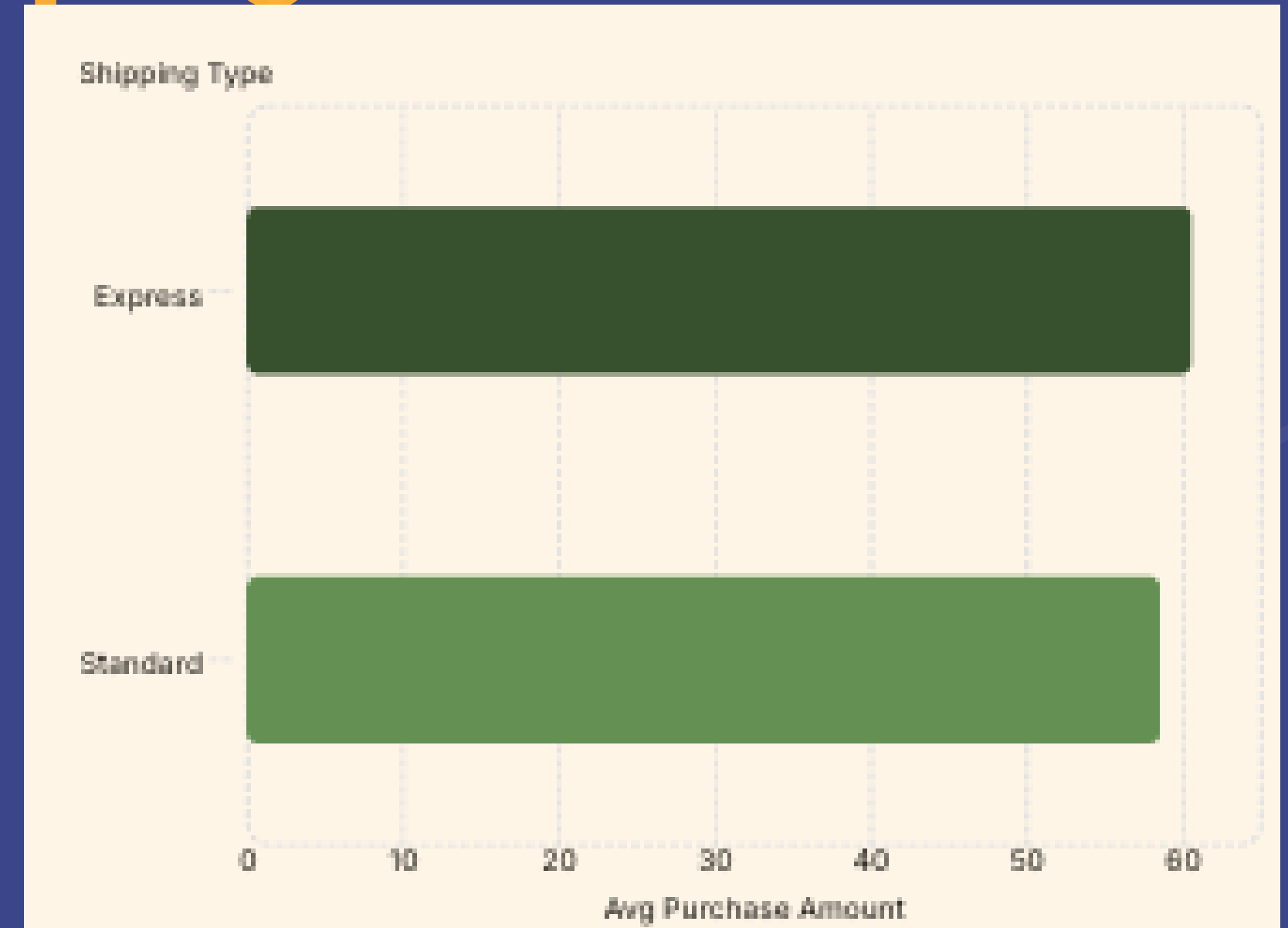
Key Finding Male customers generated more than double the revenue of female customers — \$157,890 vs. \$75,191. This significant gap suggests a key segment to target for growth.

Male revenue is 2.1× higher than female revenue across 3,900 transactions



Discount Users & Shipping Preferences

Customer ID	Purchase Amount
2	64
3	73
4	90
7	85
9	97
12	68



Several customers who applied discounts spent above the average purchase amount:

Express shipping customers spend slightly more on average (\$60.48 vs. \$58.46).

Top 5 Best-Reviewed Products

Skirt

Avg Rating: 3.78

Hat

Avg Rating: 3.80

 Boots

Avg Rating: 3.82

 Sandals

Avg Rating: 3.84

 Gloves

Avg Rating: 3.86

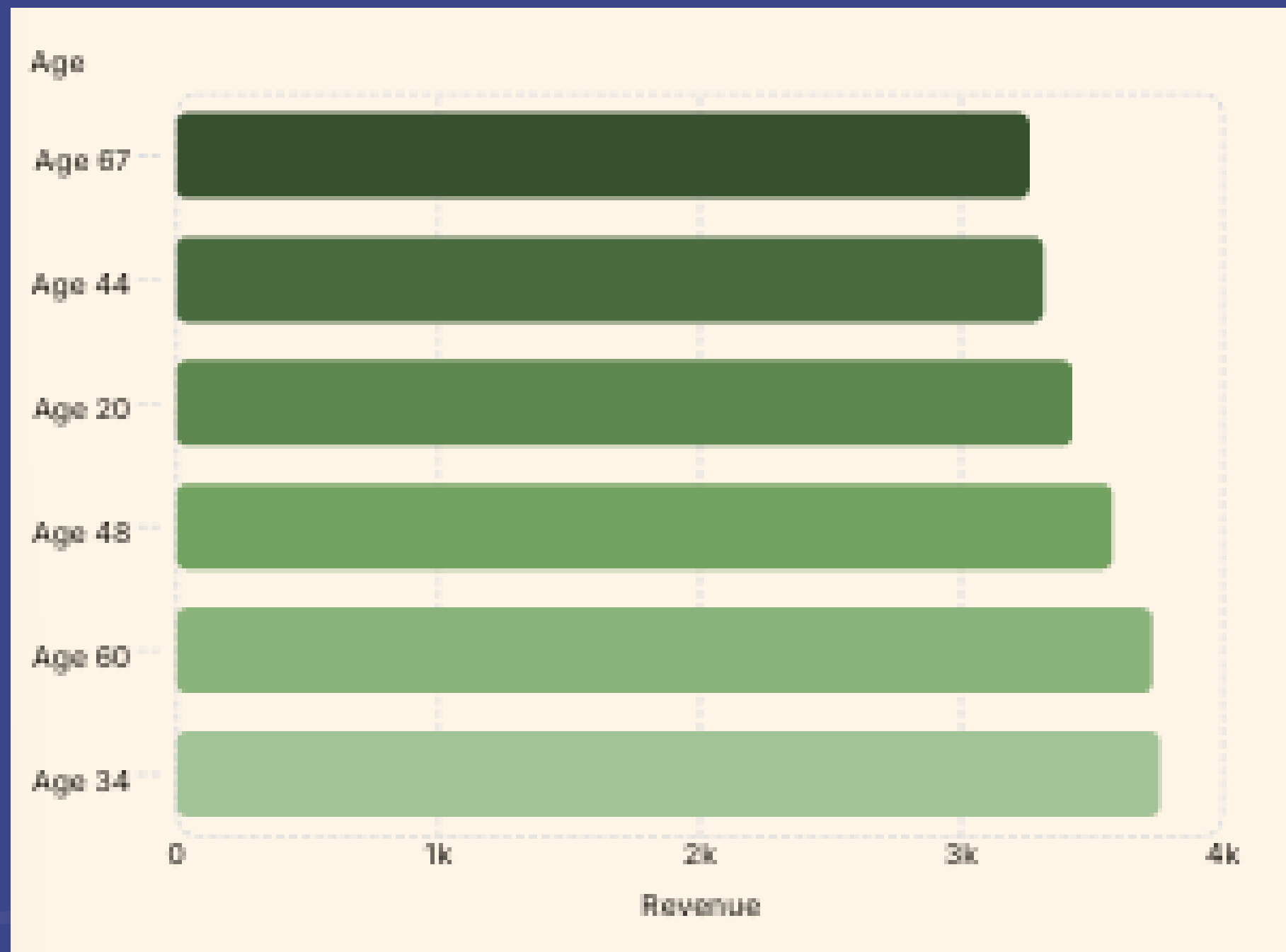
Subscriber vs. Non-Subscriber Behavior

Key Insights

- 2,847 non-subscribers generated \$170,436 in revenue
- 1,053 subscribers generated \$62,645 in revenue
- Average spend is nearly identical (~\$59.50 vs. ~\$59.87)
- Non-subscribers represent the larger customer base



Revenue Contribution by Age



Key Finding

Revenue is relatively evenly distributed across age groups, with Age 34 contributing the highest (\$3,759) and Age 67 the lowest (\$3,258).

Young adults and middle-aged customers lead in both revenue and customer count, making them the most valuable demographic segments.

Power BI Dashboard Highlights

Revenue by Category

Clothing leads (~\$0.08M), followed by Accessories, Footwear & Outerwear

Subscription Status

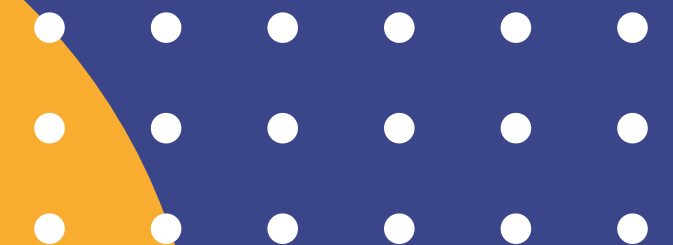
92.7% non-subscribers vs. 7.3% subscribers

Revenue by Age Group

Young adults generate the highest revenue (~\$55K), followed by middle-aged, adult, and senior

Avg Purchase Amount

\$59.76 average across 3,900 transactions with 3.75 avg review rating



CUSTOMER BEHAVIER DASHBOARD

subscription_stat...

No

Yes

gender

Female

Male

category

Accessories

Clothing

Footwear

Outerwear

shipping_type

All

3.75

avg review rating

3900

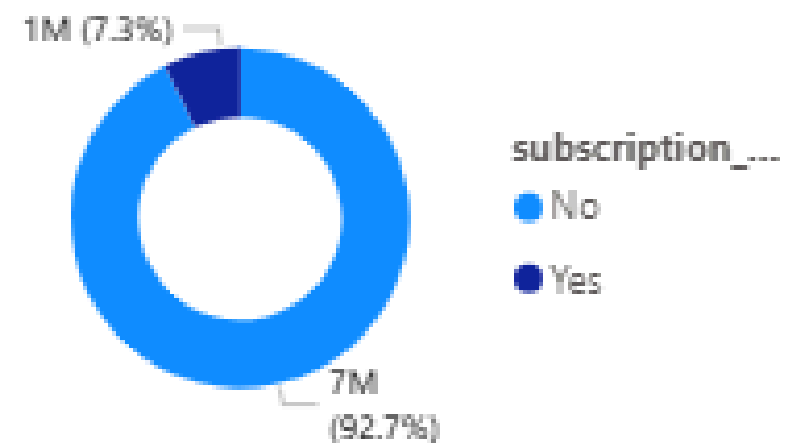
new customer id

59.76

avg purchase amount

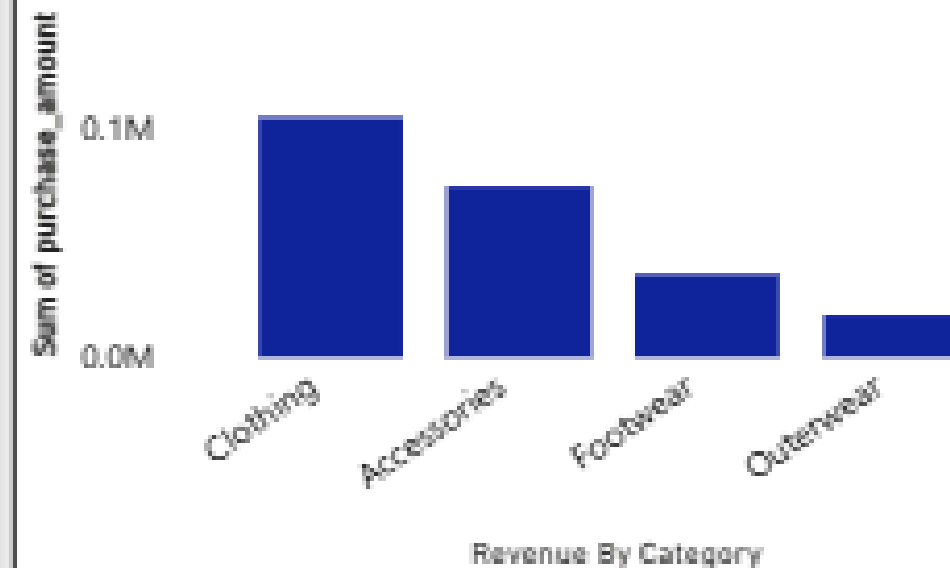
% of customers by subscription Status

Sum of customer_id by subscription_status



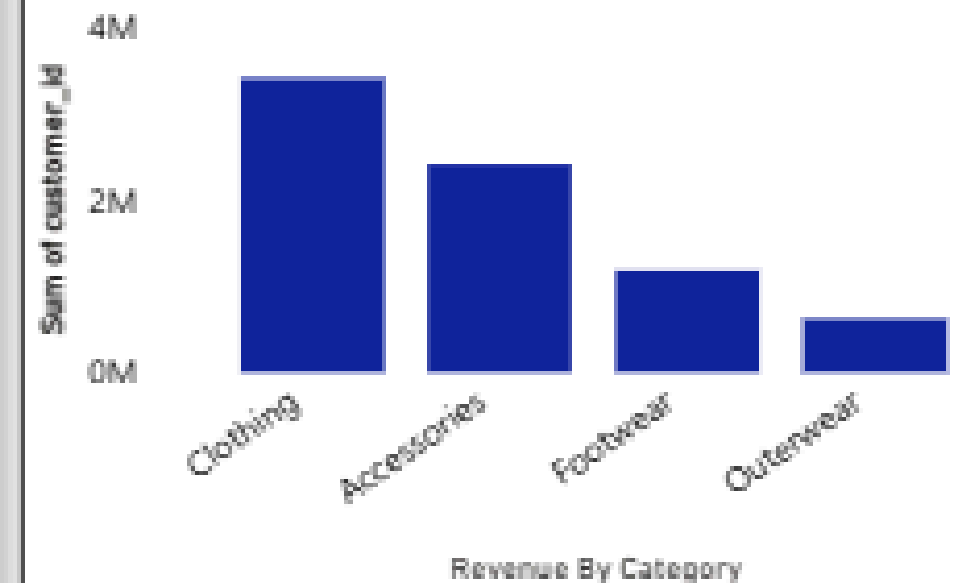
Revenue By Catagory

Sum of purchase_amount by category



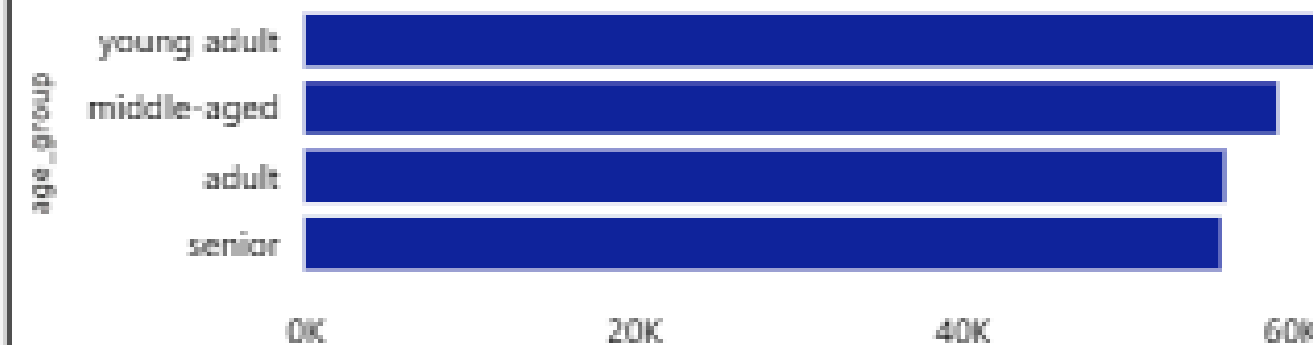
Sales By Catagory

Sum of customer_id by category



Revenue By Age Group

Sum of purchase_amount by age_group



Sales By Age Group

Sum of customer_id by age_group

